

A Microscopic Derivation of the Cross-Correlation Kernel

In this appendix, we provide the rigorous microscopic derivation of the cross-coupling term introduced phenomenologically in Eq. (2) of the main text. We demonstrate how the inter-bath coupling $\hat{H}_{B_1 B_2}$ naturally generates a cross-spectral density $J_{12}(\omega)$, and we numerically validate that the phenomenological correlation coefficient $\rho \approx 0.855$ (extracted in Paper I) corresponds to a physically realistic inter-bath coupling strength.

A.1 Total Hamiltonian and Mass-Weighted Coordinates

We consider a quantum system S coupled to two bosonic baths, B_1 and B_2 , which are themselves mutually coupled. The total Hamiltonian is given by the generalized Caldeira–Leggett model:

$$\hat{H}_{\text{total}} = \hat{H}_S + \hat{H}_{B_1} + \hat{H}_{B_2} + \hat{H}_{SB_1} + \hat{H}_{SB_2} + \hat{H}_{B_1 B_2}, \quad (1)$$

where the bath Hamiltonians and their mutual coupling are:

$$\hat{H}_{B_i} = \sum_k \left(\frac{\hat{p}_{k,i}^2}{2m_{k,i}} + \frac{1}{2} m_{k,i} \omega_{k,i}^2 \hat{q}_{k,i}^2 \right), \quad i \in \{1, 2\}, \quad (2)$$

$$\hat{H}_{B_1 B_2} = \sum_{k,j} \lambda_{kj} \hat{q}_{k,1} \hat{q}_{j,2}. \quad (3)$$

To decouple the kinetic energy we introduce mass-weighted coordinates and momenta, $\tilde{\mathbf{Q}} = \mathbf{M}^{1/2} \mathbf{Q}$ and $\tilde{\mathbf{P}} = \mathbf{M}^{-1/2} \mathbf{P}$, where $\mathbf{M}$ is the block-diagonal mass matrix. The bath Hamiltonian then takes the compact form:

$$\hat{H}_{\text{baths}} = \frac{1}{2} \tilde{\mathbf{P}}^T \tilde{\mathbf{P}} + \frac{1}{2} \tilde{\mathbf{Q}}^T \tilde{\mathbf{K}} \tilde{\mathbf{Q}}, \quad (4)$$

where $\tilde{\mathbf{K}} = \mathbf{M}^{-1/2} \mathbf{K} \mathbf{M}^{-1/2}$ is the symmetric, mass-weighted stiffness matrix with elements $\tilde{\lambda}_{kj} = \lambda_{kj} / \sqrt{m_{k,1} m_{j,2}}$.

A.2 Canonical Transformation to Normal Modes

Since $\tilde{\mathbf{K}}$ is a real symmetric matrix it can be diagonalised by an orthogonal matrix $\mathbf{O}$ ($\mathbf{O}^T \mathbf{O} = \mathbf{I}$):

$$\mathbf{O}^T \tilde{\mathbf{K}} \mathbf{O} = \mathbf{\Omega}_{\text{diag}}^2 = \text{diag}(\tilde{\omega}_1^2, \tilde{\omega}_2^2, \dots, \tilde{\omega}_N^2), \quad (5)$$

where $\tilde{\omega}_\alpha$ are the eigenfrequencies of the *normal modes*. Defining the normal-mode coordinates $\mathbf{X} = \mathbf{O}^T \tilde{\mathbf{Q}}$ (a canonical transformation) fully decouples the bath Hamiltonian:

$$\hat{H}_{\text{baths}} = \sum_{\alpha=1}^N \left(\frac{\hat{P}_{X,\alpha}^2}{2} + \frac{1}{2} \tilde{\omega}_\alpha^2 \hat{X}_\alpha^2 \right). \quad (6)$$

The system–bath coupling $\hat{H}_{SB} = \hat{X}_S \otimes \mathbf{g}^T \mathbf{Q}$ transforms to $\hat{H}_{SB} = \hat{X}_S \otimes \sum_\alpha \tilde{g}_\alpha \hat{X}_\alpha$, with effective normal-mode coupling:

$$\tilde{g}_\alpha = (\mathbf{O}^T \mathbf{M}^{-1/2} \mathbf{g})_\alpha. \quad (7)$$

Partitioning $\mathbf{O}$ into blocks reflecting each original bath,

$$\mathbf{O} = \begin{pmatrix} \mathbf{O}_{11} & \mathbf{O}_{12} \\ \mathbf{O}_{21} & \mathbf{O}_{22} \end{pmatrix}, \quad (8)$$

the effective coupling splits as $\tilde{g}_\alpha = \tilde{g}_{1,\alpha} + \tilde{g}_{2,\alpha}$, where:

$$\tilde{g}_{1,\alpha} = \sum_k (\mathbf{O}_{11})_{k\alpha} \frac{g_{k,1}}{\sqrt{m_{k,1}}}, \quad \tilde{g}_{2,\alpha} = \sum_j (\mathbf{O}_{21})_{j\alpha} \frac{g_{j,2}}{\sqrt{m_{j,2}}}. \quad (9)$$

A.3 Explicit Definition of the Cross-Spectral Density

The total spectral density of the decoupled normal modes reads

$$J_{\text{total}}(\omega) = \frac{\pi}{2} \sum_\alpha \frac{\tilde{g}_\alpha^2}{\tilde{\omega}_\alpha} \delta(\omega - \tilde{\omega}_\alpha).$$

Expanding $\tilde{g}_\alpha^2 = (\tilde{g}_{1,\alpha} + \tilde{g}_{2,\alpha})^2$ decomposes the spectral density into three contributions:

$$J_{\text{total}}(\omega) = J_{11}(\omega) + J_{22}(\omega) + 2 J_{12}(\omega), \quad (10)$$

where the **cross-spectral density** is defined explicitly by:

$$J_{12}(\omega) = \frac{\pi}{2} \sum_\alpha \frac{\tilde{g}_{1,\alpha} \tilde{g}_{2,\alpha}}{\tilde{\omega}_\alpha} \delta(\omega - \tilde{\omega}_\alpha). \quad (11)$$

When $\lambda_{kj} = 0$ the matrix $\mathbf{O}$ is block-diagonal, so $\mathbf{O}_{21} = 0$ and therefore $J_{12}(\omega) = 0$ identically. Hence $J_{12}(\omega) \neq 0$ is the direct mathematical signature of bath entanglement mediated by $\hat{H}_{B_1 B_2}$.

A.4 Cauchy–Schwarz Inequality and Numerical Validation

From Eq. (11), the Cauchy–Schwarz inequality for finite sums guarantees, for every ω ,

$$|J_{12}(\omega)|^2 \leq J_{11}(\omega) J_{22}(\omega). \quad (12)$$

This bound allows us to define a normalised, frequency-dependent cross-correlation coefficient:

$$\rho(\omega) = \frac{J_{12}(\omega)}{\sqrt{J_{11}(\omega) J_{22}(\omega)}}, \quad |\rho(\omega)| \leq 1. \quad (13)$$

Numerical link to Paper I. In Paper I a phenomenological analysis of the decoherence exponent yielded an effective inter-bath correlation coefficient $\rho \approx 0.855$. To validate that this value is physically attainable, we constructed a minimal model of two Ohmic–Drude baths with $N_1 = N_2 = 2$ modes each and performed a parameter sweep over the coupling strength λ (with $\lambda_{kj} = \lambda \sqrt{\omega_{k,1} \omega_{j,2}}$). Numerical diagonalisation shows that $\rho_{\text{peak}} = \max_\omega |\rho(\omega)|$ increases monotonically with λ while strictly satisfying Eq. (12). An optimal coupling of $\lambda_{\text{opt}} \approx 0.095$ reproduces $\rho_{\text{peak}} \approx 0.855$. This corresponds to a weak-to-moderate inter-bath coupling, $\|\hat{H}_{B_1 B_2}\|/\|\hat{H}_{B_i}\| \lesssim 12\%$, which justifies the microscopic derivation via canonical transformation without requiring a full non-perturbative treatment, and confirms that the correlation $\rho \approx 0.855$ extracted in Paper I is not a mathematical artefact but reflects a realistic physical coupling between the environmental modes.